

Challenge 02 - Hash cracking and analyzing a certificate

1. Capture and save a PCAP file of the TLS connection establishment (handshake) from your browser to the found website you've found by cracking the hash. Apply a capture filter(!) to limit capturing only to traffic between your browser and the target system. Record the capture filter you applied in your report.

Websites found using this command in Kali (*john hash.txt --format=whirlpool --wordlist=cybersecurity_sites.txt*): **hackerone.com**



Analyzing_cert.pcap

2. Analyze the connection to the website and answer to the following questions in a written report. Make sure to include a proof of evidence (e.g., screenshot, used tool) for each answer in your report. Hint: Think about an easy way to get the information. It does not necessarily have to be extracted from the PCAP file.
- For which period (start and expiration date) is the certificate valid?



Zertifikatsinformationen

Dieses Zertifikat ist für folgende Zwecke beabsichtigt:

- Garantiert dem Remotecomputer Ihre Identität
- Garantiert die Identität eines Remotecomputers
- 2.16.840.1.114412.2.1
- 2.23.140.1.1

* Weitere Infos finden Sie in den Angaben der Zertifizierungsstelle.

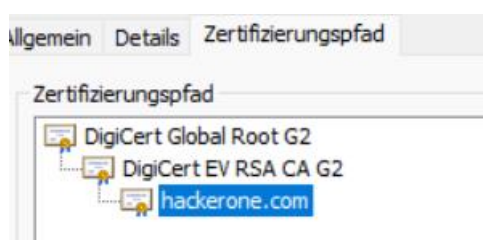
Ausgestellt für: hackerone.com

Ausgestellt von: DigiCert EV RSA CA G2

Gültig ab 07.03.2025 **bis** 08.04.2026

Start: 07.03.2025 till 08.04.2026

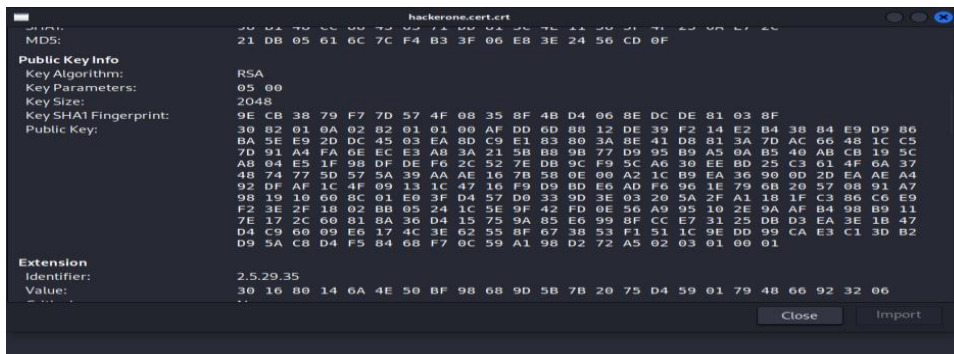
What is the name of the organization which issued the certificate?



Root CA: DigiCert Global Root G2

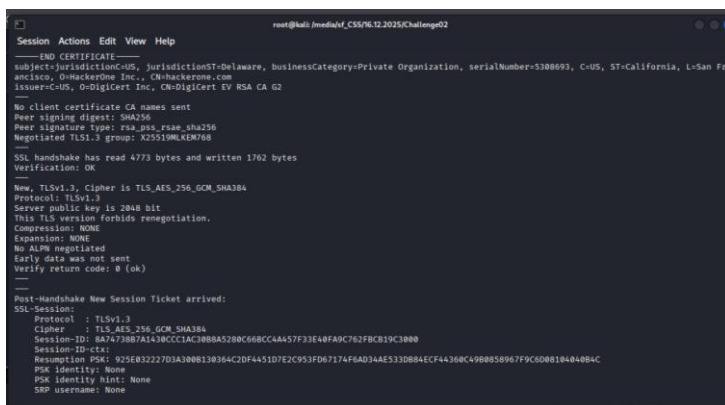
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- Which encryption standard was used to create the public key of the certificate and what is the length of the key?



RSA Encryption was used. The key size is 2048

- What TLS version was negotiated between your browser and the server for communication?



Tp2 was being displayed in the web browser,

Checked again in the terminal using this command: `# openssl s_client -connect hackerone.com:443`.

How many different encryption standards did your browser offer the server to encrypt the data?

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✓ Cipher Suites (16 suites)

Cipher Suite: Reserved (GREASE) (0x9a9a)
Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)
Cipher Suite: TLS_AES_256_GCM_SHA384 (0x1302)
Cipher Suite: TLS_CHACHA20_POLY1305_SHA256 (0x1303)
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 (0xc02b)
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (0xc02f)
Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 (0xc02c)
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (0xc030)
Cipher Suite: TLS_ECDHE_ECDSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca9)
Cipher Suite: TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (0xcca8)
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA (0xc013)
Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc014)
Cipher Suite: TLS_RSA_WITH_AES_128_GCM_SHA256 (0x009c)
Cipher Suite: TLS_RSA_WITH_AES_256_GCM_SHA384 (0x009d)
Cipher Suite: TLS_RSA_WITH_AES_128_CBC_SHA (0x002f)
Cipher Suite: TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)

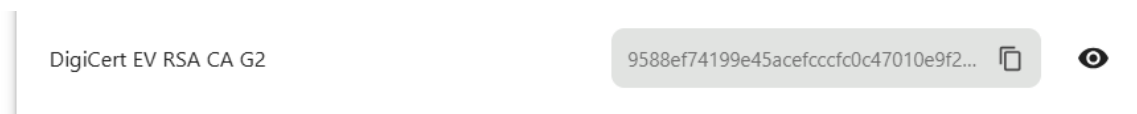
I have found the cipher suites in the Wireshark, under the Transport Layer Security in the cipher suites.

Which encryption standard and key length was finally used to encrypt the data?

Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)

This one is also under the cipher suite

How and where could your browser determine if the certificate was already revoked before its expiration date?



1. Viewed the certificate in your browser by identifying the Issuer Name (the intermediate or root CA that issued the certificate).
2. Checked the root CA in the browser's trusted store. I confirmed that the root CA exists in the browser's trusted CA store.
 - This step ensures the certificate is cryptographically signed by a trusted authority.
3. Using OpenSSL, I extracted the OCSP URL directly from the certificate:

```
# openssl x509 -in hackerone.cert.crt -noout -ocsp_uri
```

4. Also found out serial number:

```
# openssl x509 -in hackerone.cert.crt -noout -serial
```

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5. And then I used both serial number and the URL to see if its revoked or not:

```
# openssl ocsp -issuer issuer.crt -cert hackerone.cert.crt -url
```

<http://ocsp.digicert.com>

```
(root@kali)-[/media/sf_CSS/16.12.2025/Challenge02]
# openssl ocsp -issuer issuer.crt -cert hackerone.cert.crt -url http://ocsp.digicert.com
WARNING: no nonce in response
Response verify OK
hackerone.cert.crt: good
    This Update: Dec 18 09:57:00 2025 GMT
    Next Update: Dec 25 08:57:00 2025 GMT
(root@kali)-[/media/sf_CSS/16.12.2025/Challenge02]
```

Found the result. Its not revoked before the expiration date.